Kartik Madan

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Education

Guru Gobind Singh Indraprastha University — New Delhi, India
Bachelor of Engineering in Electronics and Communication; GPA: 3.28 (8.2/10.0) — *2016 - 2020*

Experience

BP p.l.c. — *Sep 2024 - Present*

Conversational Analytics Platform for Trading Operations — London, United Kingdom
Software Engineer

- Delivered an analytics platform for the trading desk end to end: a data-intensive React/TypeScript interface traders use daily, FastAPI backend, authentication and session management, CI/CD pipelines and AWS CDK infrastructure-as-code, cutting query turnaround from hours to seconds.
- Designed the backend services and REST APIs that let trading operators query BP's trading warehouse, orchestrating a four-stage request pipeline over enterprise datasets.
- Built an agent-orchestration layer over large language models (AWS Bedrock) that turns operators' natural-language questions into accurate, governed SQL, grounded in business rules, warehouse metadata and domain terminology.
- Engineered a query safety and validation layer (read-only enforcement, row limits, injection protection) to guarantee the correctness and security of every response.
- Built automated unit, integration and end-to-end tests across frontend and backend behind a CI coverage gate, along with an AI evaluation harness scoring text-to-SQL accuracy against known cases.

Contract Value Leakage — London, United Kingdom
Software Engineer

- Built document-processing pipelines and backend services reconciling ~15K supplier invoices/month against agreed price books.
- Developed extraction and validation services spanning 200+ supplier templates and 50+ invoice formats at 95%+ accuracy.
- Implemented an invoice-to-contract matching and anomaly-detection engine, reducing undetected contract leakage by ~40%.

Wild Bean Cafe — London, United Kingdom
Full Stack Engineer and Project Lead

- Led development of a cloud-native pricing, sales and inventory platform for self-service Micro Market units across 100+ BP forecourt retailers in the UK and Poland.
- Partnered with POS, SAP and platform teams to build the enterprise integration and analytics workflows connecting in-store systems to cloud data platforms.
- Added natural-language analytics letting retail managers query operational data directly.
- Contributed to a 148% improvement in dealer profitability through automation and reporting.

Store of Today — Gurgaon, India
Full Stack Engineer

- Built event-driven services for real-time hazard detection using computer vision across retail locations.
- Developed dynamic pricing over Electronic Shelf Labels (ESLs).
- Automated AWS infrastructure and deployment pipelines with Terraform.

Infosys — *Nov 2020 - Sep 2024*

Equinox Studio — Gurgaon, India
Full Stack Engineer

- Built a low-code and no-code digital commerce platform used by global clients (Kraft Heinz, Shutterfly, Herbalife, Nu Skin), contributing 22% of the team's revenue.

Projects & Open Source

- **Energy Trading:** Market-participation platform for India's power-trading ecosystem, with ML-driven forecasting.
- **Price Forecasting Engine:** RNN-based deep-learning app that forecasts market price movements and streams predictions into live trading platforms.
- **Microsoft Agent Framework:** Open-source contributor to Microsoft's framework for building and orchestrating agent workflows.
- **ContractIQ:** System to create and analyze legal contracts, built on LangChain.

Skills

- **Languages:** JavaScript, TypeScript, Python, C++, SQL, C#, Java
- **Frontend:** React, Next.js, Vue, HTML and CSS
- **Backend:** FastAPI, Node.js, REST APIs
- **Cloud & DevOps:** AWS, GCP, Terraform, Docker, CI/CD (Azure DevOps)
- **Data:** PostgreSQL, DynamoDB, OpenSearch, BigQuery, analytical and columnar stores, SQL and NoSQL
- **AI/ML:** LLMs, RAG, agent orchestration, embeddings and vector search

Achievements

- **Codeforces:** World Rank 625, National Rank 37 (of 7,585) at best finish.
- **LeetCode:** Knight, 1,854 contest rating.
- **Hackathons:** Winner, Smart India Hackathon 2019; Winner, Barco Geekathon 2018.
- **Infosys RISE Awards:** Business Ninja (Q3 FY24), Tech Wizard (Q4 FY23).

Publications

- **Patent (US 2026/0141595, published):** System and method for identifying and replacing proprietary images with synthetically generated images.
- **Springer, ICICC 2020:** Development of a real-time pollution monitoring system for green auditing.